

Pedestrian Route Choice Simulation

Using Persona-informed AI Agents

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URBAN PLANNING, GENERATIVE AGENTS, OUTDOOR THERMAL COMFORT



BACKGROUND

Pedestrian route-choice modelling underpins walkability, accessibility, and outdoor thermal comfort research. Yet most approaches rely on representative agents with utility- maximising rules, averaging out the individual differences that shape real walking experience.

GAP

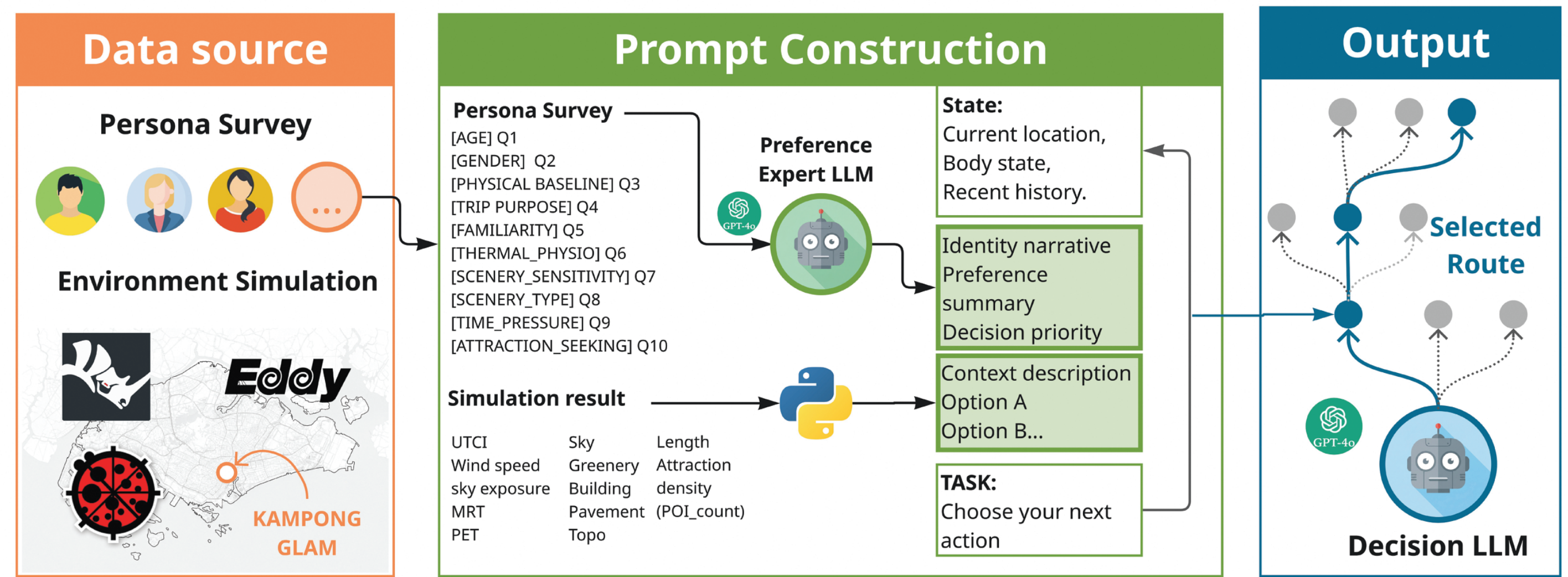
- Conventional ABM averages out the individual differences that shape real walking experience.
- Recent LLM-agent work treats weather as a scene descriptor and lets the model infer how the body responds.
- Thermal load and fatigue — central to street-scale comfort in the tropics — are not modelled quantitatively.

AIM

To support more responsive urban design by simulating how different pedestrians experience the same street under real microclimate conditions — and to deliver this through PersonaWalk, an interactive platform for designers and city managers.

METHODS

A 10-question survey defines each persona — age, physical baseline, trip purpose, thermal sensitivity, and aesthetic preferences. Microclimate (UTCI, wind, MRT) and street features are simulated in Rhino with Ladybug and Eddy3D. At every intersection, the agent's persona, body state (thermal load, fatigue), and surroundings are passed to an LLM, which chooses the next street and narrates the experience.



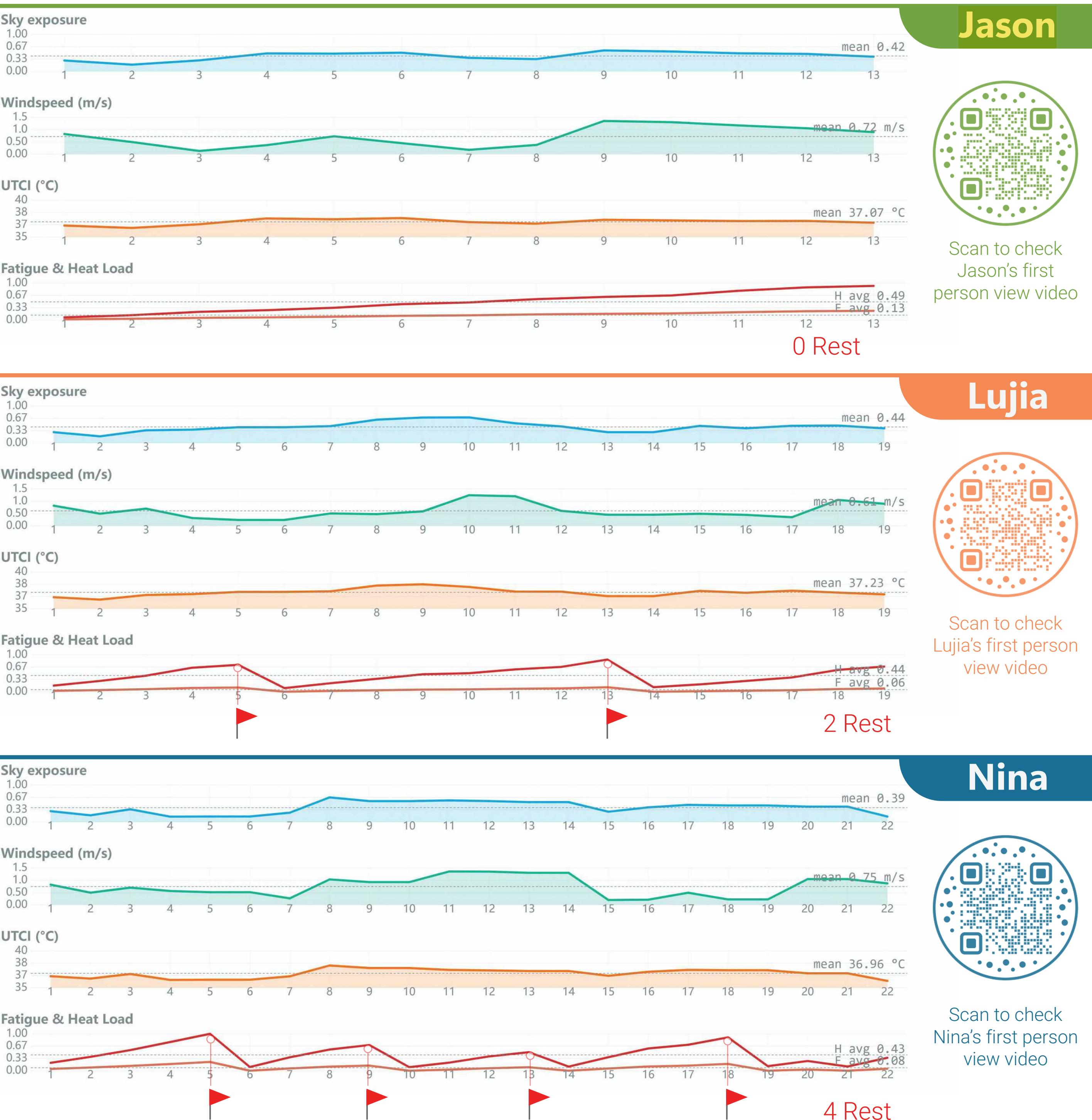
FINDINGS

58 personas were simulated walking between the same start and end points in Kampong Glam, with three highlighted here. Given identical origins and destinations, personas chose different routes, walked different distances, and rested different numbers of times. Heat sensitivity, age, and scenery preference each pulled route choice away from the shortest path which conventional models cannot capture.



KEY DISCUSSION POINTS

What this method offers—Each route comes with readable reasoning and feedback, making persona-specific design trade-offs visible.
What this method is not—LLM personas are not real users. PersonaWalk supports early design exploration, it does not replace engagement with real communities.

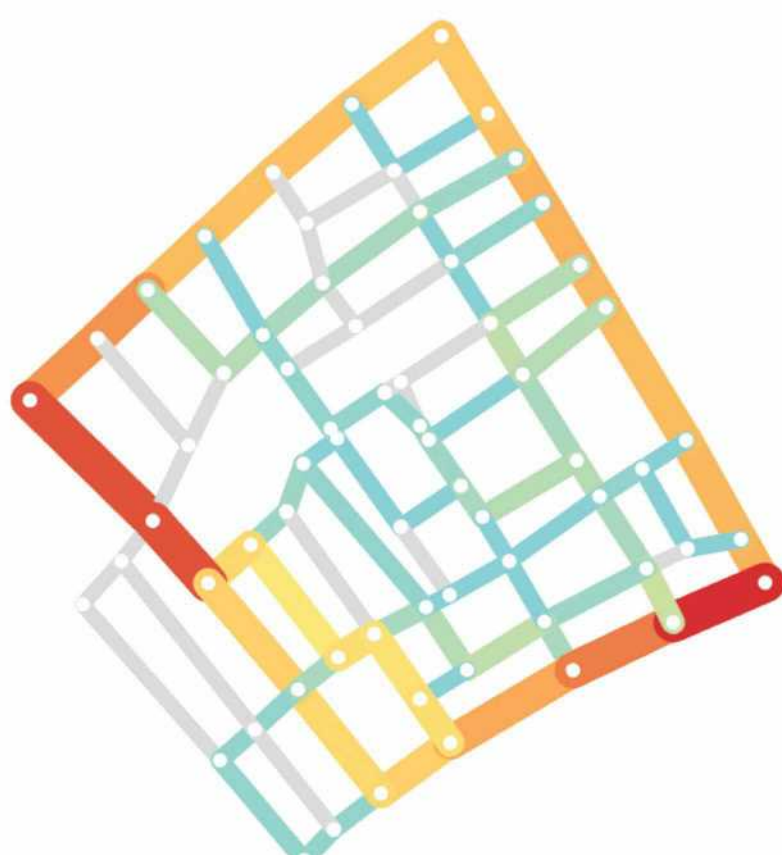


CONCLUSIONS

This work delivers two outputs: a persona-informed LLM-agent framework that integrates microclimate and physiological state, and PersonaWalk, an interactive platform that makes persona-specific routes, reasoning, and feedback inspectable. Together they give designers and city managers a way to test whose street it is — before a single brick is laid.

Future Application

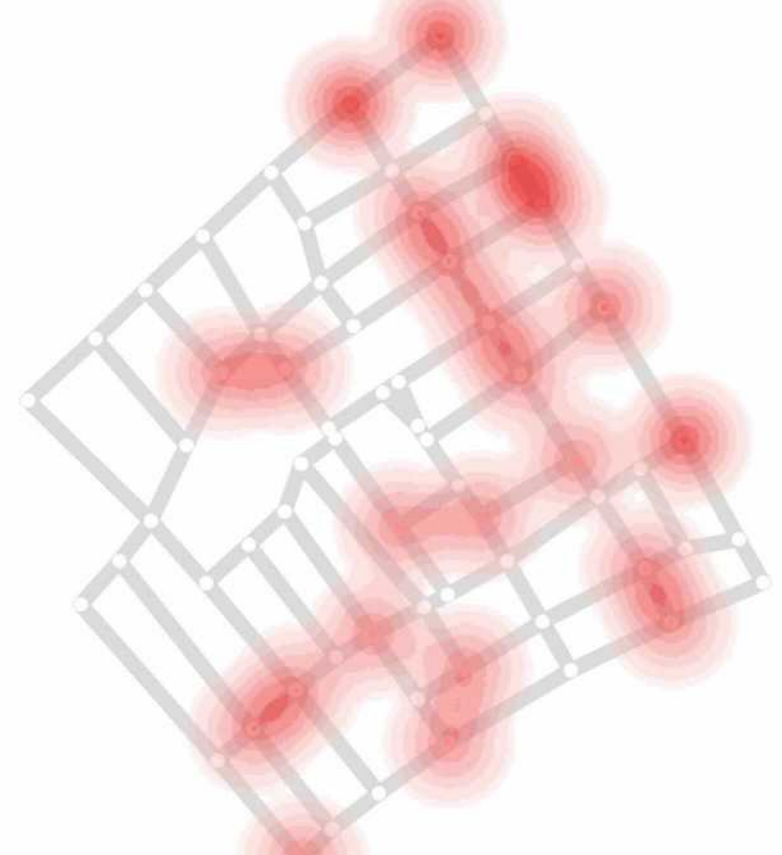
Identify popular streets at population scale



Map heat-stress hotspots across the network



Locate rest-spot priorities for seating and shade



Pre-test design interventions before construction



Contact

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Acknowledgements

The study is supported by the CoolNUS-BEAM initiative

Funded by

